

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice. (2 points each)

Amphipods	Cnidocyte	Shoal
CaCO ₃	Cryptic coloration	SiO ₂
Cellulose	Isopod	Spicules
Choanocyte	Notochord	Spongin
Copepods	Octopus	Squid
Countershading	Osculum	Tentacle
Cnidae	Ostium	Trochophore
Cnidaria	School	Trophosome

1. _____ Stinging cells in coral polyps.
2. _____ Laterally (side-to-side) flattened arthropods.
3. _____ Water enters sponges through this hole.
4. _____ Mollusk that lacks any type of shell.
5. _____ Fish in an undirected group for social reasons.
6. _____ Larval form of some annelids.
7. _____ The foraminiferan shell is composed of this chemical compound.

(6 pts) 8. Briefly describe two possible functions of parapodia in polychaete worms.

- (6 pts) 9. List the scientific names for the three phyla of macroalgae. For each phylum, list the proper term that identifies the primary photosynthetic pigment that determines its overall color.

Phylum

Pigment

_____	_____
_____	_____
_____	_____

- (4 pts) 10. With regard to energy and carbon sources, describe the one difference and the one similarity between a photoautotroph and a chemoautotroph.

- (5 pts) 11. Briefly explain why small zooplankton like copepods must remain near the ocean surface even though they are not photosynthetic.

- (10 pts) 12. On screen is a picture of a juvenile blueface angelfish. Describe at least two different ways in which the coloration and pattern help to protect this fish from predation on the coral reef.

(10 pts) 13. Many nudibranchs are brightly colored and therefore highly visible to potential predators. The colors are hypothesized to serve as warning coloration. To support this hypothesis, briefly describe one defense mechanism that nudibranchs use to guard against predation.

(10 pts) 14. A new group of deep-sea echinoderms was discovered in 1986. These deep-sea animals, called sea daisies, are flat and round, and look very much like sea stars without arms. List three adult characters and one juvenile character that sea daisies should possess to confirm them as echinoderms.

- (10 pts) 15. Despite the low Reynold's number for many marine planktonic organisms, they still tend to sink. Describe three physical and biological mechanisms that help planktonic organisms remain at or near the ocean surface. Do not include the Reynold's number.

2 pts extra credit: Taylor did not know one newly discovered homologous structure that unites urochordates and chordates. Now he knows. Let's find out if you know, too. Name the structure. Do not list the other four that he already knew or he may subtract points.

1 pt extra credit: Taylor did not know what adult freshwater lampreys eat. (Did he know *anything*?) Now he knows. Let's find out if you know, too. What do adult freshwater lampreys eat? Use one word and one word only to answer this question. He will have read enough of your other answers by now.